

Cloud-Based Hostel Mess Food Ordering & Billing System

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Problem & Need

Problems:

- Manual token collection causes long queues.
- Monthly billing handwritten → errors occur.
- Students waste time waiting.

Solutions:

- Cloud-based ordering + auto billing reduces delay and errors.

System Description

- Students place orders for breakfast, lunch, or dinner using a mobile/web app.
- Orders go to a cloud server, kitchen staff view orders, and monthly bills are auto-generated.

Users & Cloud Requirements

Users:

- Students – order food & view bill
- Mess Staff – view orders, update menu
- Admin – track billing reports

Cloud Needs:

- Compute – order processing & bill calculation
- Storage – save menu, order history, billing
- Networking – remote access via internet

Cloud Characteristics Mapping

- On-Demand Self-Service – students order anytime without permission.
- Scalability – system expands when 100+ orders come at peak time.
- Measured Service – billing is based on orders
→ automatic measurement.

Role of Data Centre

- Stores all mess data: menu, orders, payment history.
- Provides backup if system crashes.
- Ensures 24×7 availability.
- Keeps billing and logs securely.

Architecture Diagram



Challenges & Solution

Challenges:

- Internet required for ordering
- System may overload during peak time

Solutions:

- Cloud auto-scaling and offline token fallback mode

Conclusion

- Cloud enables automated food ordering and accurate billing.
- It saves time, reduces queues, improves student convenience,
- and allows mess admin to access reports anywhere.